



What is Symposium?

- Extended Rust toolchain for agentic development

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- Extended Rust toolchain for agentic development
- Marquee feature:
 - Installing skills, hooks, etc based on your dependencies

Member of the Rust Foundation's Innovation Lab

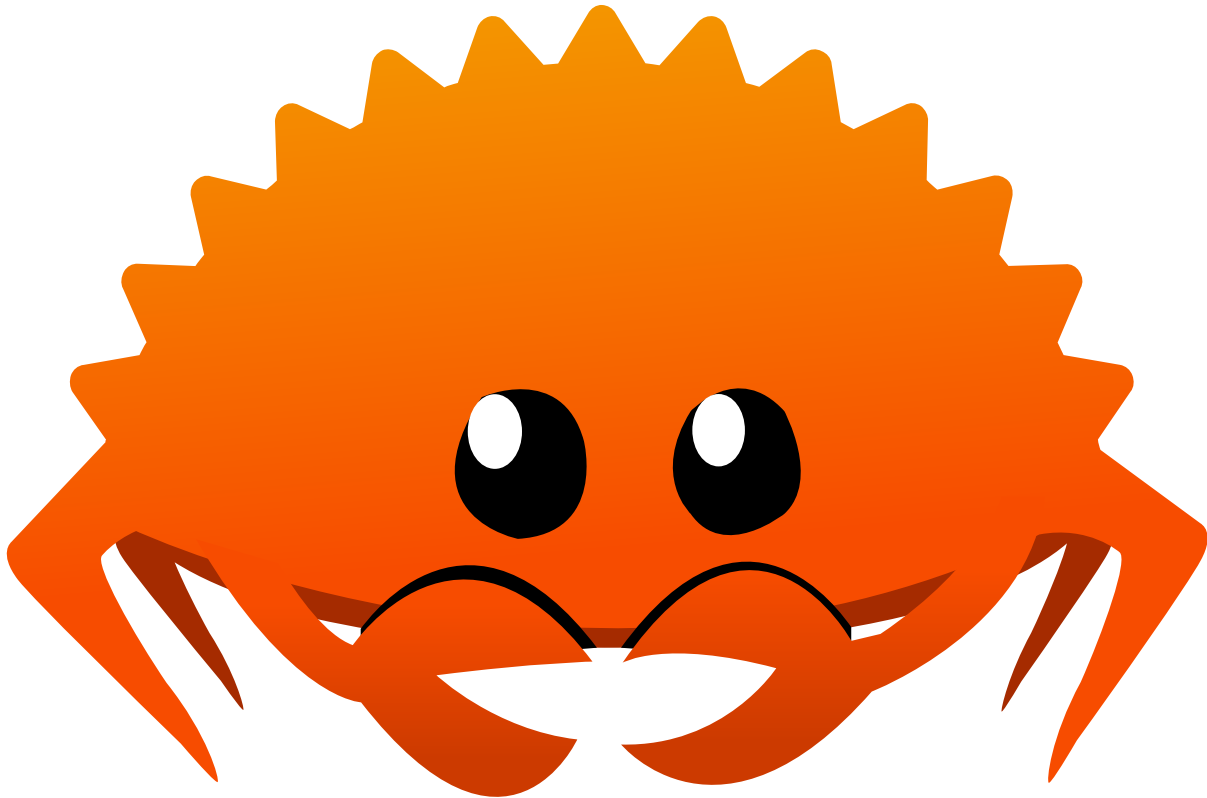
Welcoming Symposium to the Rust Innovation Lab

April 21, 2026 / Rust Foundation Team

Why Symposium?

Who am I?

One of the lead designers of Rust



Who am I?

Senior Principal Engineer at Amazon



Not heard of Rust?

Low-level enough for a kernel...

usable enough for an application



Low-level enough for a kernel?

Low-level enough for a kernel?

Rust Goes Mainstream in the Linux Kernel

Rust is no longer experimental. It's now as acceptable as C in the Linux kernel.

Dec 10th, 2025 12:00pm by [Steven J. Vaughan-Nichols](#)

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Rust in the Windows Kernel: Microsoft's 2026 Safety Push

Microsoft is rewriting Windows kernel components in Rust. What's been rewritten, the security rationale, and what it means for Rust developers in 2026.



Max Wells

Published: Mar 27, 2026 · Updated: May 4, 2026

High-level enough for an application?

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TUI oscilloscope written using ratatui

Rust has been on a roll lately

But two main things hold us back:

- "Rewrites are expensive, interop is hard"

Rust has been on a roll lately

But two main things hold us back:

- "Rewrites are expensive, interop is hard"
- "Learning curve"

Lately... things look a bit different

// My goal is to eliminate every line of C and C++ from Microsoft by 2030. Our strategy is to combine AI *and* Algorithms to **rewrite Microsoft's largest codebases**. Our North Star is "1 engineer, 1 month, 1 million lines of code".

— Distinguished Engineer at Microsoft, [talking about a research project](#) (emphasis mine)

Really different

// If you've already tried Rust and found the learning curve too steep, give it another try with Claude Code or Codex as your pair programmer. **The experience is different when you have an AI that can navigate ownership and borrowing patterns** while you focus on building things.

The tools finally catching up to the promise of the language.

— Tigran Bayburtsyan, "[Coding Rust with Claude Code and Codex](#)" (emphasis mine)

When using agents...

Pick a language, libraries, etc based not on the code you have now,
but based on the **code you want to maintain going forward**.

When your slogan is "a stitch in time, saves nine" in an agentic world...

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Rust right now

Rust + LLM = dream-team?

- Guardrails

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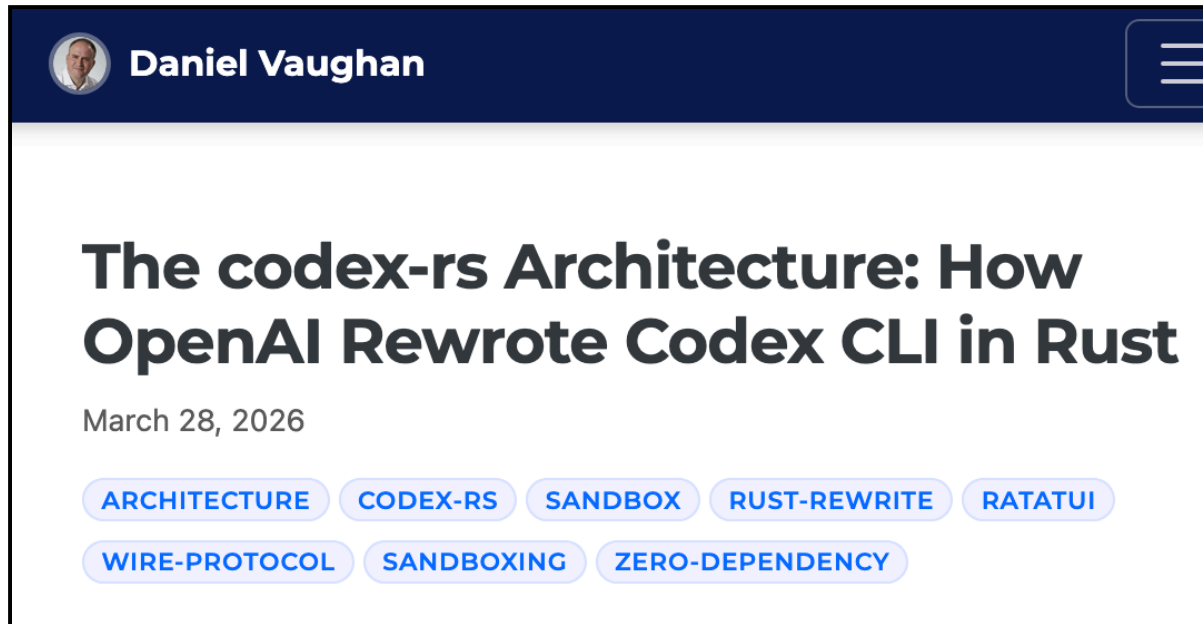
Co-founder and president of OpenAI.

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- Efficiency

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Cloudflare just got faster and more secure, powered by Rust

2025-09-26

ENGINEERING

Counter Service: How we rewrote it in Rust



Jia Long Loh · Pu Li · Muqi Li · Md Riyadh

20 Jun 2025 | 13 min read

Abstract

The Integrity Data Platform (IDP) team decided to rewrite our microservices in Rust. It resulted in [70% infra](#) savings and fewer its pitfalls. This article will elaborate on:

2x Performance, \$300k Savings: A Case Study in Rewriting a Critical Service in Rust

A Tale of Scale: When Your Service Hits a Wall

In the world of high-traffic systems, success often brings its own set of challenges. During my internship at TikTok, one of our core payment services—a reliable workhorse built in Go—began showing signs of strain. As TikTok LIVE's user base grew, the CPU

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- Guardrails
- Versatility
- Efficiency
- Investment in error messages

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Error messages?

```
Compiling playground v0.0.1 (/playground)
error[E0308]: mismatched types
  --> src/main.rs:6:15
   |
6  |     print_row(row);
   |     ^^^^^^   ^^^ expected `&str`, found `String`
   |
   | arguments to this function are incorrect
   |
note: function defined here
  --> src/main.rs:1:4
   |
1  | fn print_row(s: &str) {
   |     ^^^^^^^^^^
help: consider borrowing here
6  |     print_row(&row);
   |               +
```

For more information about this error, try ``rustc --explain E0308``.

Error messages?

```
Compiling playground v0.0.1 (/playground)
error[E0308]: mismatched The base error
  --> src/main.rs:6:15
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The base error

Needed context

How to fix

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So... Rust + agents are good

But could they be better?

Example: Toasty

Toasty, an async ORM for Rust, is now on crates.io

April 3, 2026

Toasty is an asynchronous ORM for the Rust programming language that prioritizes ease of use. Toasty supports both SQL and NoSQL databases. Today, on the NoSQL front, only DynamoDB works, but I'm hoping to get more support in the coming months.

Here is how you might use it:

```
#[derive(Debug, toasty::Model)]
struct User {
    #[key]
    #[auto]
    id: u64,

    name: String,

    #[unique]
    email: String,

    #[has_many]
    todos: toasty::HasMany<Todo>,
}
```

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Example: Using Toasty

```
// Create a user, the ID is not specified as it is set automatically  
toasty::create!(User {  
  name: "John Doe",  
  email: "john@example.com",  
}).exec(&mut db).await?;  
  
// Find the user by email  
let user = User::get_by_email(&mut db, "john@example.com").await?;
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Macros and custom syntax

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New methods

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New methods



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- **Versatility:** High-level and declarative
- **Efficiency:** Compiles to efficient code

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Type-safe, efficient

```
// Find the user by email
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Claude doesn't even know it exists

"What library do you recommend for working with sqlite in Rust?"

My usual recommendation:

Situation	Reach for
CLI tool / simple script	<code>rusqlite</code>
Async service (Tokio)	<code>sqlx</code>
Complex domain model, migrations-first	<code>diesel</code>
Async + ORM ergonomics	<code>sea-orm</code>

For the record, these too are all excellent libraries!

Claude, can you use toasty?

can you write me an execute that uses the Rust crate toasty to create a local database with all jpegs in the current directory and their file size

Claude, can you use toasty?

Synthesized Toasty framework patterns and architected JPEG database solution ▾

🕒 Let me check the toasty crate to understand its API before writing code.

🌐 toasty Rust crate ORM database getting started 8 results

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- 📄 Announcing Toasty, an async ORM for Rust | Lobsters lobste.rs
- 📦 toasty - crates.io: Rust Package Registry crates.io
- 📄 toasty - Rust docs.rs docs.rs

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That's a lot of tokens...

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- 🔗 GitHub - tokio-rs/toasty: An async ORM for Rust · GitHub [github.com](#)
- 📄 Announcing Toasty, an async ORM for Rust | Lobsters [lobste.rs](#)
- 📦 toasty - crates.io: Rust Package Registry [crates.io](#)
- 📄 toasty - Rust [docs.rs](#)

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Now I have everything I need. Here's the full example using Toasty 0.3 with the `#[derive(toasty::Model)]` macro API (no schema file — that was the old approach):

Cargo.toml

toml

```
[package]
name = "jpeg-index"
version = "0.1.0"
edition = "2021"

[dependencies]
toasty = { version = "0.3", features = ["sqlite"] }
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Toasty on crates.io

 **crates.io**

toasty



 Browse All Crates |  Log in with GitHub

Search Results for 'toasty'

Displaying 1-10 of 19 total results

Sort by  Relevance ▾

toasty v0.5.0

An async ORM for Rust supporting SQL and NoSQL databases

[Homepage](#) [Documentation](#) [Repository](#)

↓ All-Time: 4,304

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The world is moving faster than ever

Training data can't keep up

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Rust 2024 hit stable Feb 2025

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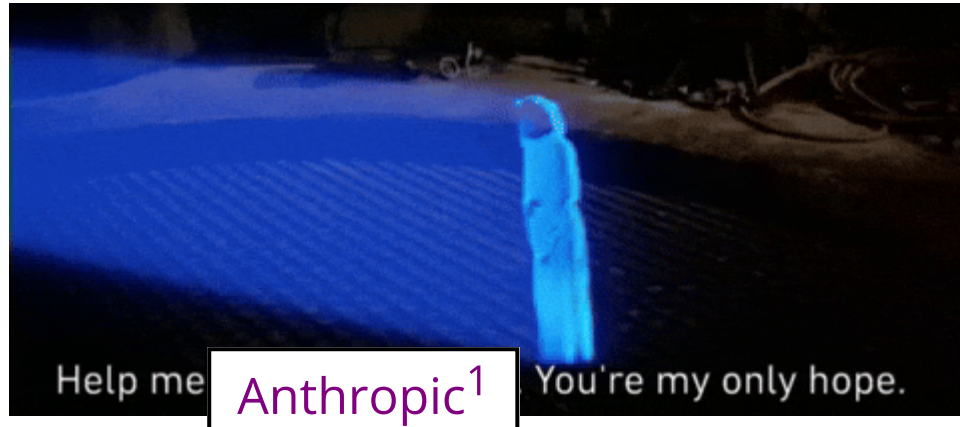
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Most frustrating thing of all?

Most frustrating thing of all?

...the Rust org cannot help



¹ I should say: I tried that toasty example twice, and the second time, Claude did use Toasty 0.5. But it still made a Rust crate in the 2021 edition.

..and this is where Symposium comes in



Install symposium

```
$ cargo install symposium # or cargo binstall!
```

Install symposium

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```
$ cargo agents init
```

Setting up symposium for your user account.

Which agents do you use? (space to select, enter to confirm):

```
> [x] Claude Code
```

```
    [x] Codex CLI
```

```
    [ ] GitHub Copilot
```

```
    [ ] Gemini CLI
```

```
    [ ] Goose
```

```
    [x] Kiro
```

```
    [ ] OpenCode
```

Symposium gives general guidance

- General Rust guidance, e.g.,
 - Use Rust 2024
 - Use `cargo add` to add crates so you get the latest version
 - Run `cargo fmt` after making edits

Symposium lets you write skills for crates

name	toasty-guidance
description	Guidance for using the Toasty async ORM crate — schema definition, CRUD, relations, queries, and transactions
crates	toasty
activation	always

Toasty is an async ORM for Rust supporting SQL (SQLite, PostgreSQL, MySQL) and NoSQL (DynamoDB). It prioritizes type safety and leans into each database's capabilities rather than hiding them.

Schema definition

Define models with `#[derive(toasty::Model)]`. Mark the primary key with `#[key]` and auto-generated fields with `#[auto]` (auto-increment for integers, UUID v7 for `uuid::Uuid`).

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Symposium installs skills automatically

- Install hooks:
 - After every tool use, we check for new relevant skills and install them
 - **Result:** After agent runs `cargo add toasty`, it knows how to use it

Symposium provides APIs to help agents

```
> cargo agents crate-info serde
```

```
Crate: serde
```

```
Version: 1.0.228
```

```
Source: /Users/nikomat/.cargo/registry/src/index.crates.io/serde-1.0.228
```

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> cargo agents crate info serde
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Version: 1.0.228

Source: /Users/nikomat/.cargo/registry/src/index.crates.io/serde-1.0.228

 **Correct version used by your code**

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```
> cargo agents crate-info serde
```

```
Crate: serde
```

```
Version: 1.0.228
```

```
Source: /Users/nikomat/.cargo/registry/src/index.crates.io/serde-1.0.228
```



Let the agent browse full sources

Symposium aims to make agents extensible

"Everybody has something unique to offer"



Symposium provides for interoperability

As a library author, write one set of extensions that work across agents...

Symposium provides for interoperability

As a library author, write one set of extensions that work across agents...

...good luck with that.

Even AGENTS.md is not universally supported

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AGENTS.md

A simple, open format for guiding coding agents, used by over 60k open-source projects.

Think of AGENTS.md as a **README for agents**: a dedicated, predictable place to provide the context and instructions to help AI coding agents work on your project.

[Explore Examples](#)

[View on GitHub](#)

AGENTS.md

Setup commands

- Install deps: `pnpm install`
- Start dev server: `pnpm dev`
- Run tests: `pnpm test`

Code style

- TypeScript strict mode
- Single quotes, no semicolons
- Use functional patterns where possible

One AGENTS.md works across many agents

Your agent definitions are compatible with a growing ecosystem of AI coding agents and tools:



Amp



Factory



Windsurf
from Cognition



Phoenix

Agents



Ona



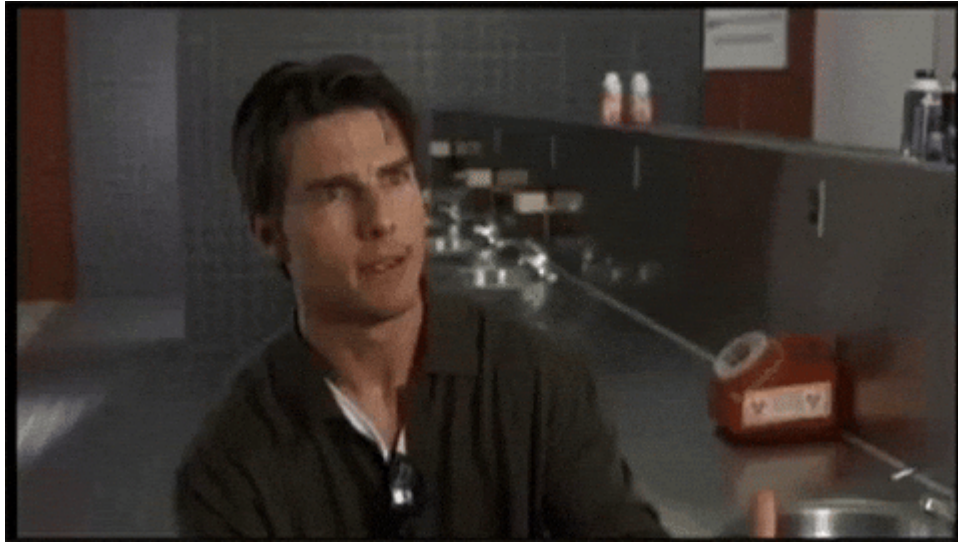
opencode



Devin
from Cognition

[View all supported agents](#)

Niko to vendors...



Symposium does the best we can

- Library (crate) provides extensions
 - MCP servers
 - Skills
 - Hooks

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Symposium does the best we can

- Library (crate) provides extensions
 - MCP servers
 - Skills
 - Hooks
- User picks agent
- Symposium adapts
 - Load skills into the right directories for the agent user chose
 - Converts between hook formats

Conclusions

- If you want something to be amazing:
 - Unleash the ecosystem!

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- Symposium aims to take a good thing (Rust + LLMs) and make it **better**:
 - Skills and extensions based on what you use
 - Up-to-date guidance

Conclusions

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Use Rust? Try it now!

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$ cargo binstall symposium
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```
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